

F24-125-D-AdvertEase

Project Team

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Chapter 1

Introduction

The purpose of this project scope document is to clearly define the objectives our projects, its scope, the main motivation behind developing an automated system for the business industry to create poster and video advertisement at just a single click. This document deeply explains the problem statement and the solution we proposed to solve this problem. We have also enlisted the main modules and features of the project. AdvertEase, our solution, aims to revolutionize the way traditional advertisements are created and make it much more efficient, easy and budget friendly. It allows the businesses to simply enter the URL of the product to generate a prompt based customized poster advertisement as well as different video advertisements with human avatar along with enhanced emotion for voiceover and gestures. Following is our basic prototype which we want to create :

1.1 Problem Statement

Small or even middle-tier furniture businesses often struggle with creating compelling video and poster ads for their products. Creating video/poster ads is always a hassle and usually a difficult task, as companies either need to hire freelancers or have specific teams for such tasks further increasing the cost. Hence the task to market product becomes difficult, costly and most importantly time-consuming. With the passage of time increased usage of social media has made it very important for the companies to create video content to attract the audience and boost sales. Also due to the lack of the tools having the enhanced emotion voice overs with human avatars the AI generated video content is much more robotic and less attractive. The lack of the budget friendly and compelling solutions makes it very difficult for these businesses to compete in an increasingly digital marketplace which in result hinders their growth and performance.



Figure 1.1: Poster Advertisement

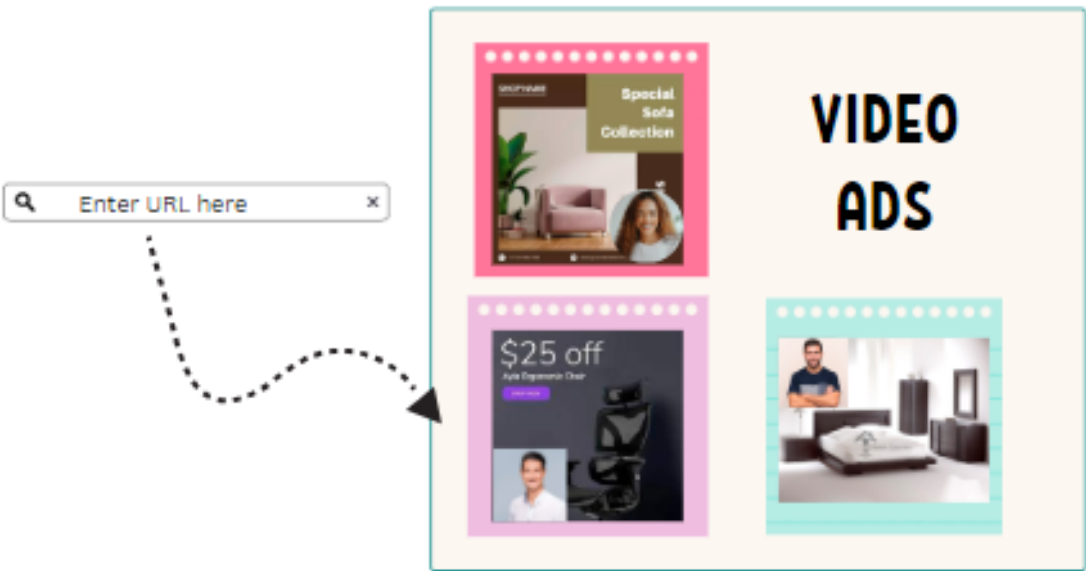


Figure 1.2: Video Advertisement

1.2 Scope

The project focuses on building AI-generated poster and video ads for furniture industry. Initially, a product link is given to our application, which will automatically scrape product descriptions from the web, preprocess them, and feed it to an LLM, which will generate the poster advertisement description. The description will be given to an image generator model, which will generate the poster. customer can further tweak their advertisement by giving prompt to their model regarding changes in the poster. For video ads, a description from an LLM will be created and fed into the video generator model, which will generate a video . The video will have a human avatar with enhanced emotions and gestures explaining the product description. There will be both a male as well as a female avatar trained for the advertisements. Our application will be web based and will be deployed on cloud (AWS) for reliability and availability. However, the app does not scrap data from all websites but rather a selected few, the video ads created will not be dynamic; instead, templates will be used, however, the content of the videos will be dynamic and change with each product

1.3 Modules

1.3.1 Module 1 : AI generated Poster advertisements

This module will be responsible for creating unique and attractive poster ads for our products. The ads will include tags like sale, buy one get one free, etc to enhance customer engagement. Users can further tweak this poster by providing feedback to the model and the poster will be regenerated, reflecting the required changes

1. Automated web scraping
2. Text pre-processing
3. LLM script generation for poster generation
4. Image generation by models
5. Prompt based image editing

1.3.2 Module 2 : AI generated Video advertisements

This module will be responsible for creating engaging videos for the product. We will be using in-painting/Gen AI techniques to map our product into the video. The video will

also consist of an avatar which will be explaining the product features as the video goes on. That avatar will be a key point in the video as it will persuade the viewers to buy that product.

1. LLM script generation for video generation
2. Video generation by video inpainting / Camera Control /Motion Control techniques
3. Avatar generation for video
4. Creating templates for the Generated product videos and avatar

1.3.3 Module 3 : Website Development

Our application will be deployed as a website so we aim to develop a website using MERN stack. Our website would be easy to use and have a visually attractive appeal to attract customers

1. Frontend development
2. Backend development
3. CI/CD pipeline

1.3.4 Module 4 : Cloud Deployment

Our application will be deployed on the cloud for further reliability and durability. By cloud deployment, our application would be fault tolerant and less prone to common on premise deployment issues

1. Deployment on AWS

1.4 User Classes and Characteristics

Identify the various user classes that you anticipate will use this product, and describe their pertinent characteristics.

User class	Description
End User	An individual or company trying to create advertisements for their goods or services is the end user. Product URLs will be entered, preferences will be chosen, and the created advertising will be downloaded or shared. Because users might not be highly proficient in technology, the system must be easy to use. The users, who may come from a variety of industries like retail or e-commerce, demand a fast and easy ad generating procedure.
Admin	The administrator oversees the platform's operation, handles user management, keeps an eye on system performance, and makes sure the AI tool for creating ads is operating at peak efficiency. The administrator is also capable of managing problems with system updates, authentication, and user registration. In the event of a system failure or to update algorithms for improved efficiency, they might step in.
Marketing experts	The task of improving the scripts, avatars, and templates utilised in ad generating falls to a marketing specialist. They evaluate the efficacy of different advertising tactics and offer suggestions for raising the quality of voiceovers, video material, and poster creation. Their feedback aids in the optimisation of ad formats for particular target markets.
Developers	The backend of the system is maintained and updated by the developer, who also makes sure that upgrades, bug fixes, and new features are applied properly. They are in charge of AWS deployment, cloud services, AI modules like Llama3 and Stable Diffusion, and system integration. Their tight collaboration with the admin guarantees a seamless user experience.

Chapter 2

Project Requirements

2.1 Use-case/Event Response Table/Storyboarding

Following is the use case diagram of our system. Our system includes three major actors

- User - Ones who uses the system to generate the advertisement
- System - AI system which generated things on the backend
- Admin - One who is responsible for managing the users and generating reports

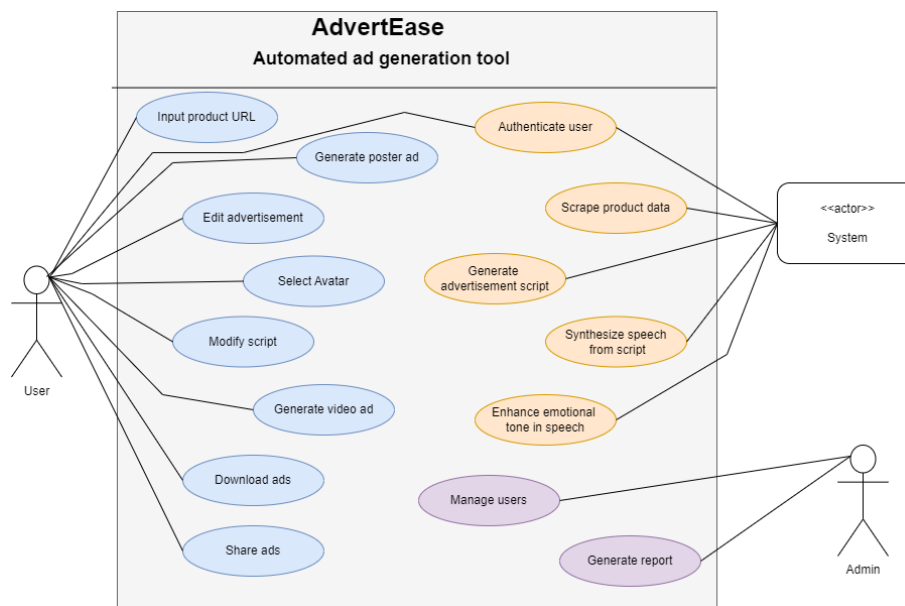


Figure 2.1: Usecase Diagram

2.1.1 Authenticate User

Use Case Name	Authenticate User
Scope	Authentication for system access
Level	User-goal
Actor	User, Admin
Stakeholders	System Admin, User
Preconditions	User is registered; system login page is accessible
Main Success Scenario	1. The user inputs their login credentials. 2. The system checks credentials. 3. The system provides access to the dashboard.
Postconditions	User or admin is logged into the system

Table 2.1: Authenticate User

2.1.2 Input Product URL

Use Case Name	Submit Product URL
Scope	Submit URL for data scraping
Level	User-goal
Actor	User
Stakeholders	System Admin, User
Preconditions	User is authenticated
Main Success Scenario	1. User enters the product URL. 2. The system accepts the URL. 3. The system validates a successful submission.
Postconditions	URL is submitted; system ready to scrape data

Table 2.2: Input Product URL

2.1.3 Scrape Product Data

Use Case Name	Extract Product Data
Scope	Scrape data from product URL
Level	Subfunction
Actor	System
Stakeholders	User, Product Owner
Preconditions	Product URL is submitted
Main Success Scenario	1. The system collects product photos, descriptions, and features from the URL. 2. Data is parsed and saved.
Postconditions	Data is available for generating ads

Table 2.3: Scrape Product Data

2.1.4 Generate Poster Advertisements

Use Case Name	Create Poster Advertisements
Scope	Generate static image ads
Level	User-goal
Actor	System
Stakeholders	User
Preconditions	Product data is scraped
Main Success Scenario	1. The system analyses product data. 2. The system generates a poster ad 3. An advertisement is displayed to the user.
Postconditions	Poster ad is ready for download or modification

Table 2.4: Generate Poster Advertisements

2.1.5 Generate Advertisement Script

Use Case Name	Generate Advertisement Script
Scope	Create text script for video ad
Level	Subfunction
Actor	System
Stakeholders	User
Preconditions	Product data is scraped
Main Success Scenario	1. System creates text based on the product description. 2. Script is displayed to the user.
Postconditions	Script is ready for speech synthesis or editing

Table 2.5: Generate Advertisement Script

2.1.6 Edit Advertisement

Use Case Name	Modify Advertisement Content
Scope	Edit content of poster or video ad
Level	User-goal
Actor	User
Stakeholders	User
Preconditions	Ad or script is generated
Main Success Scenario	1. The user selects an ad to alter. 2. The user updates the content, graphics, or layout. 3. System updates ads
Postconditions	Advertisement is updated with new content

Table 2.6: Edit Advertisement

2.1.7 Select Avatar

Use Case Name	Select Avatar
Scope	Choose avatar for video ad
Level	User-goal
Actor	User
Stakeholders	User
Preconditions	Product data is scraped; user is logged in
Main Success Scenario	1. The user browses avatar selections. 2. The user selects an avatar. 3. The avatar is added to the video advertisement.
Postconditions	Avatar is set for the video ad

Table 2.7: Select Avatar

2.1.8 Modify Script

Use Case Name	Edit Ad Script
Scope	Modify the automatically generated script
Level	User-goal
Actor	User
Stakeholders	User
Preconditions	Script is generated
Main Success Scenario	1. The user views the created script. 2. The user changes or rewrites the script. 3. The system stores the changed script.
Postconditions	Script is updated for speech synthesis

Table 2.8: Modify Script

2.1.9 Synthesize Speech from Script

Use Case Name	Synthesize Speech from Script
Scope	Convert the edited script to speech
Level	Subfunction
Actor	System
Stakeholders	User
Preconditions	Script is ready
Main Success Scenario	1. The system processes the script. 2. System synthesises voice audio. 3. Audio is generated with proper intonation.
Postconditions	Speech audio is ready for video ad

Table 2.9: Synthesize Speech from Script

2.1.10 Enhance Emotional Tone in Speech

Use Case Name	Enhance Emotional Tone in Speech
Scope	Add emotional modulation to synthesized speech
Level	Subfunction
Actor	System
Stakeholders	User
Preconditions	Speech is synthesized
Main Success Scenario	1. The system analyses script content. 2. An emotional tone (e.g., excitement, seriousness) is used. 3. Emotionally enriched speech is produced.
Postconditions	Emotionally enhanced speech is ready

Table 2.10: Enhance Emotional Tone in Speech

2.1.11 Generate Video Advertisement

Use Case Name	Produce Video Advertisement
Scope	Generate a complete video ad
Level	User-goal
Actor	System
Stakeholders	User
Preconditions	Speech, avatar, and script are prepared
Main Success Scenario	1. The system creates the video ad 2. The final video advertisement is displayed to the user.
Postconditions	Video ad is ready for download or modification

Table 2.11: Generate Video Advertisement

2.1.12 Download Ads

Use Case Name	Download Ads
Scope	Save locally created video and poster advertisements.
Level	User-goal
Actor	User
Stakeholders	User
Preconditions	The created ads can be downloaded.
Main Success Scenario	1. The user selects the download option. 2. The user is prompted by the system to choose a download format. 3. The user decides the format to use The system stores the advertisement on the user's device in the designated format.
Postconditions	The user can now access and successfully download the advertisement.

Table 2.12: Download Ads

2.1.13 Share Ads

Use Case Name	Share Ads
Scope	Post the created advertisements on social media.
Level	User-goal
Actor	User
Stakeholders	User, Social Media Platforms
Preconditions	The created ads are prepared for sharing.
Main Success Scenario	1. The user selects "Share" from the menu. 2. The system shows sharing choices (email, social media, etc.) 3. The user chooses the preferred sharing network 4. System posts the advertisement on the chosen network.
Postconditions	Successfully shared advertisement on the selected platform

Table 2.13: Share Ads

2.1.14 Manage Users

Use Case Name	Manage Users
Scope	The admin controls user accounts in the system.
Level	Admin-goal
Actor	Admin
Stakeholders	System Admin, Users
Preconditions	The admin is authorised and has the ability to oversee users.
Main Success Scenario	1. The admin logs into the user interface. 2. The administrator sees the user registration list 3. The admin chooses whether to add, modify, or remove a user 4. The request is processed by the system, which then modifies the user database. 5. The system displays a success message to validate the action.
Postconditions	User accounts can be successfully updated or changed.

Table 2.14: Manage Users

2.1.15 Generate Reports

Use Case Name	Generate Reports
Scope	The admin creates reports on system performance and ad production.
Level	Admin-goal
Actor	Admin
Stakeholders	Admin, Management
Preconditions	The admin is authorised and able to provide reports.
Main Success Scenario	1. The administrator chooses the option for report generation. 2. The administrator defines the report's parameters (such as the date range and user activity). 3. The system gathers information and creates the report 4. The administrator looks at or gets the report
Postconditions	A generated report is available for download or review.

Table 2.15: Generate Reports

2.2 Functional Requirements

2.2.1 Authenticate User

- 1.1 The system shall provide a login page where users can enter their login information.
- 1.2 The database shall be used by the system to verify user credentials.
- 1.3 After a successful login, the system shall grant access to the dashboard.

2.2.2 Input product URL

- 2.1 Users shall be able to enter a legitimate product URL into the system.
- 2.2 The format of the URL that the user enters shall be verified by the system.
- 2.3 After the URL is approved, the system shall send a confirmation message.

2.2.3 Scrape product data

- 3.1 Product data shall be extracted by the system from the given URL.
- 3.2 In the event of a data scraping error, the system shall handle it and alert the user.
- 3.3 The data that has been scraped shall be stored by the system in an organised manner for further processing.

2.2.4 Generate poster ads

- 4.1 The system shall utilise the product data that has been scraped to produce aesthetically appealing poster adverts.
- 4.2 Users shall be able to alter the poster advertising' design aspects using the system.
- 4.3 The user shall be given a sneak peek at the generated poster advertisement by the system.

2.2.5 Generate advertisement script

- 5.1 The system shall utilise the product data to create an advertisement script.
- 5.2 Users shall be able to modify the generated advertisement script through the system.

2.2.6 Edit advertisement

- 6.1 The script and poster ad generated by the system shall be editable by users through an interface.
- 6.2 User edits to the script and advertisement shall be saved by the system.

2.2.7 Select Avatar

- 7.1 The system should let users choose a video ad avatar.
- 7.2 A library of selectable avatars shall be made available by the system.

2.2.8 Modify script

- 8.1 After the advertisement script is generated, users shall be able to edit it using the system.
- 8.2 To make sure the script stays coherent, the system shall validate the changes.

2.2.9 Synthesize Speech from Script

- 9.1 The system shall employ text-to-speech technology to translate the commercial script into audio.

2.2.10 Enhance Emotional Tone in Speech

- 10.1 In accordance with script's nature, the system shall make the synthesised speech sound more emotionally charged.

2.2.11 Generate video ads

- 11.1 To produce a video advertisement, the system shall combine the script, synthesised speech, and poster advertisement.
- 11.2 The user shall be given a sneak peek at the generated video advertisement by the system.

2.2.12 Download ads

- 12.1 Users shall be able to download the produced video advertisements and posters in a variety of formats from the system.
- 12.2 The system shall confirm that the user has successfully downloaded the advertisements.

2.2.13 Share ads

- 13.1: Users shall have the ability to share the created adverts on social media sites using options provided by the system.
- 13.2 The system shall allow sharing through email or other channels that the user specifies.

2.2.14 Manage users

- 14.1 The admin shall be able to view a list of users who have registered on the system.
- 14.2 The admin shall be able to add, modify, and remove user accounts using the system.
- 14.3 The system shall notify users when the administrator makes modifications to their accounts.

2.2.15 Generate reports

- 15.1 The system shall allow the administrator to provide reports on user behaviour and the creation of ads.
- 15.2 The system shall require customisation choices for report criteria, such as date range and user activity.
- 15.3 The admin shall be able to retrieve created reports from the system in a variety of forms (such as PDF and CSV).

2.3 Non-Functional Requirements

2.3.1 Reliability

- REL-1: During regular operation, the system must maintain an average Mean Time Between Failures (MTBF) of at least 1000 hours.
- REL-2: Should a failure occur, the system will record the error information and notify the administrator in five minutes.
- REL-3: In the event of a breakdown, the system must have automated recovery methods that enable it to function again in two minutes.
- REL-4: To guarantee data integrity, the system must regularly backup user data and configurations, preferably once every 24 hours.

2.3.2 Usability

- USE-1: The system must have an easy-to-use interface that lets users create advertising from the dashboard in only three clicks or less.

- USE-2: To help users grasp the functionality, the system will include tooltips and contextual help for all significant functions.
- USE-3: With only one click, users will be able to access previously created advertisements from their user dashboard.
- USE-4: To guarantee that all users can efficiently use the program, the system must include accessibility capabilities such keyboard navigation and screen reader compatibility.

2.3.3 Performance

- PER-1: After the user enters the necessary product URL and data, the system will create a poster advertisement in no more than five seconds.
- PER-2: Under typical circumstances, the system should be able to parse and scrape data from a product URL in 10 seconds.
- PER-3: The system must guarantee that a video ad is created in maximum of 5 minutes and the poster ads are created in 10 seconds.
- PER-4: The system must guarantee that all functional requirements are satisfied within predetermined time frames, supporting a minimum of 100 concurrent users without compromising performance.

2.3.4 Security

- SEC-1: To prevent unwanted access, the system must encrypt all sensitive user data (such as login passwords and personal information) both in transit and at rest.
- SEC-2: Role-based access control (RBAC) will be implemented by the system to limit access to various capabilities according to user roles (Admin and User).
- SEC-3: To find and fix vulnerabilities, the system must conduct routine security audits.
- SEC-4: The system must offer defence against widely-known online threats like SQL injection, XSS, and CSRF (cross-site request forgery).

2.4 Domain Model

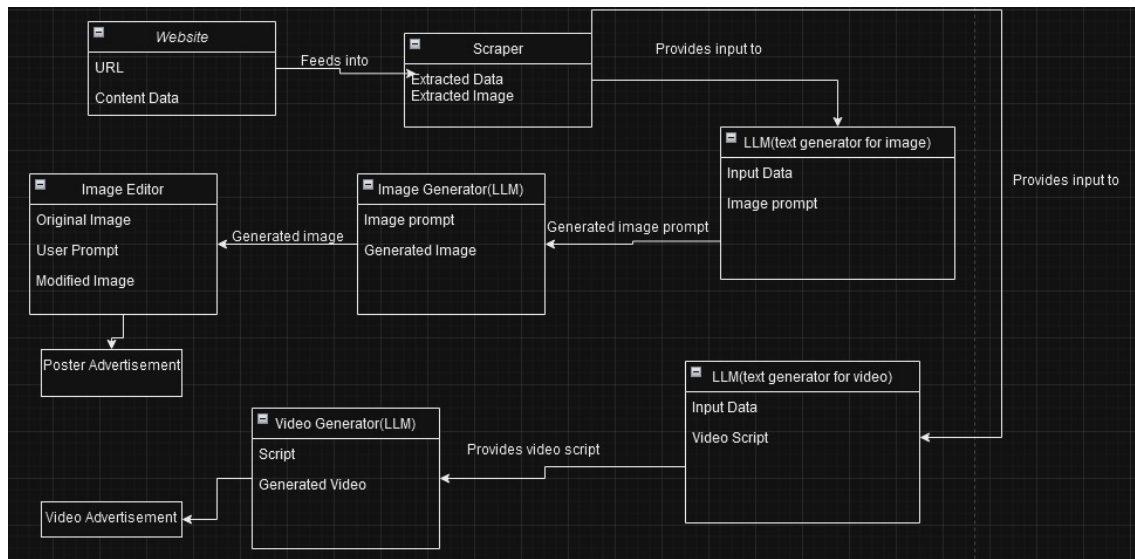


Figure 2.2: Domain Model

Chapter 3

System Overview

3.1 Architectural Design

The system is structured into four main layers:

3.1.1 Presentation Layer

- This layer includes the user interface where users interact with the system.
- Users provide input (e.g., URL for scraping, image edits, video requests).
- Flow of data occurs from this to the Application Layer.

3.1.2 Application Layer

- It is the middle layer between communication between the presentation and business logic layers.
- Receives input from the Presentation Layer (e.g., URLs, prompt editing requests).
- Passes requests to the Business Logic Layer for processing (e.g., scraping, image/video generation).
- Returns results to the Presentation Layer for display.

3.1.3 Business Logic Layer

- This layer has the core functionality of the application.

- It has all main processes such as scraping the web, LLM prompt generation, image generation, prompt-based image editing, and video generation.
- All necessary services like Scraping Service, LLM-based Text Generation, Image/Video Generators, and Prompt Editor.
- It also interacts with the Data Layer to fetch or store information (e.g., scraped data, generated media).

3.1.4 Data Layer

- It is responsible for data storage and retrieval.
- It stores scraped content, generated prompts, images, and videos.
- It works with databases or file systems to manage project data.
- It communicates with the Business Logic Layer to provide data required for image generation and video creation.

3.1.5 Flow of Information

- Data and commands flow downward from the Presentation Layer to the Data Layer for processing.
- Results and responses flow upward from the Data Layer to the Presentation Layer for user interaction.

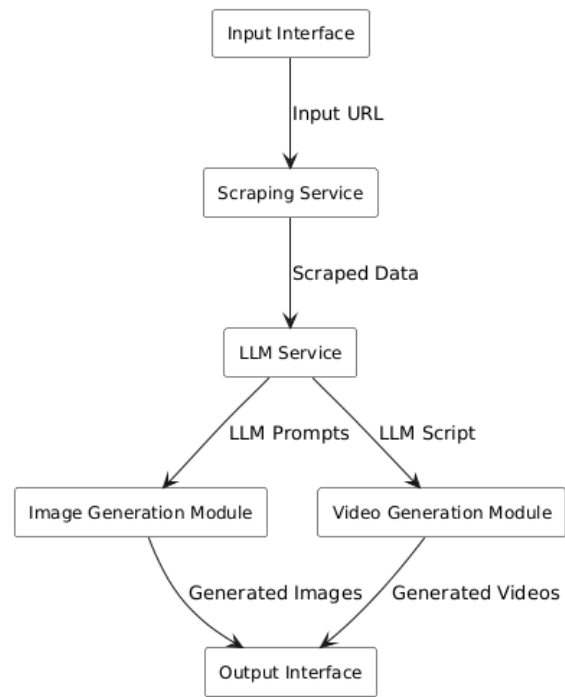


Figure 3.1: Box and Line Architecture

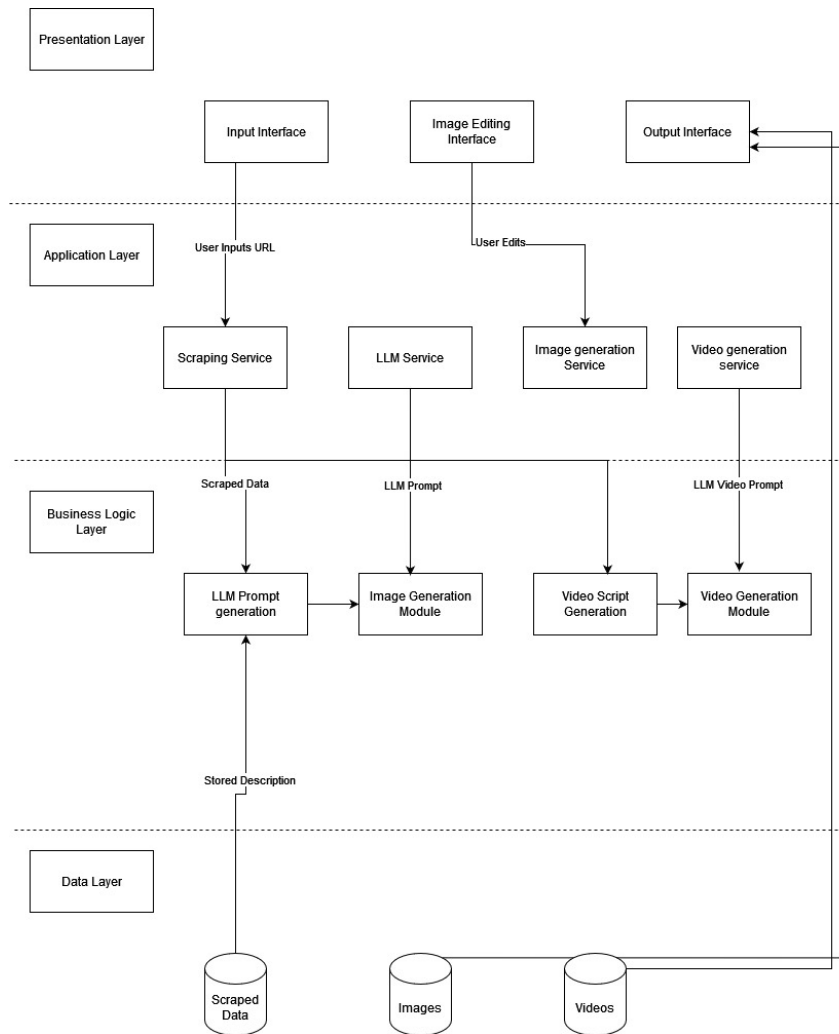


Figure 3.2: Architecture Diagram

3.2 Design Models

3.2.1 Activity Diagram

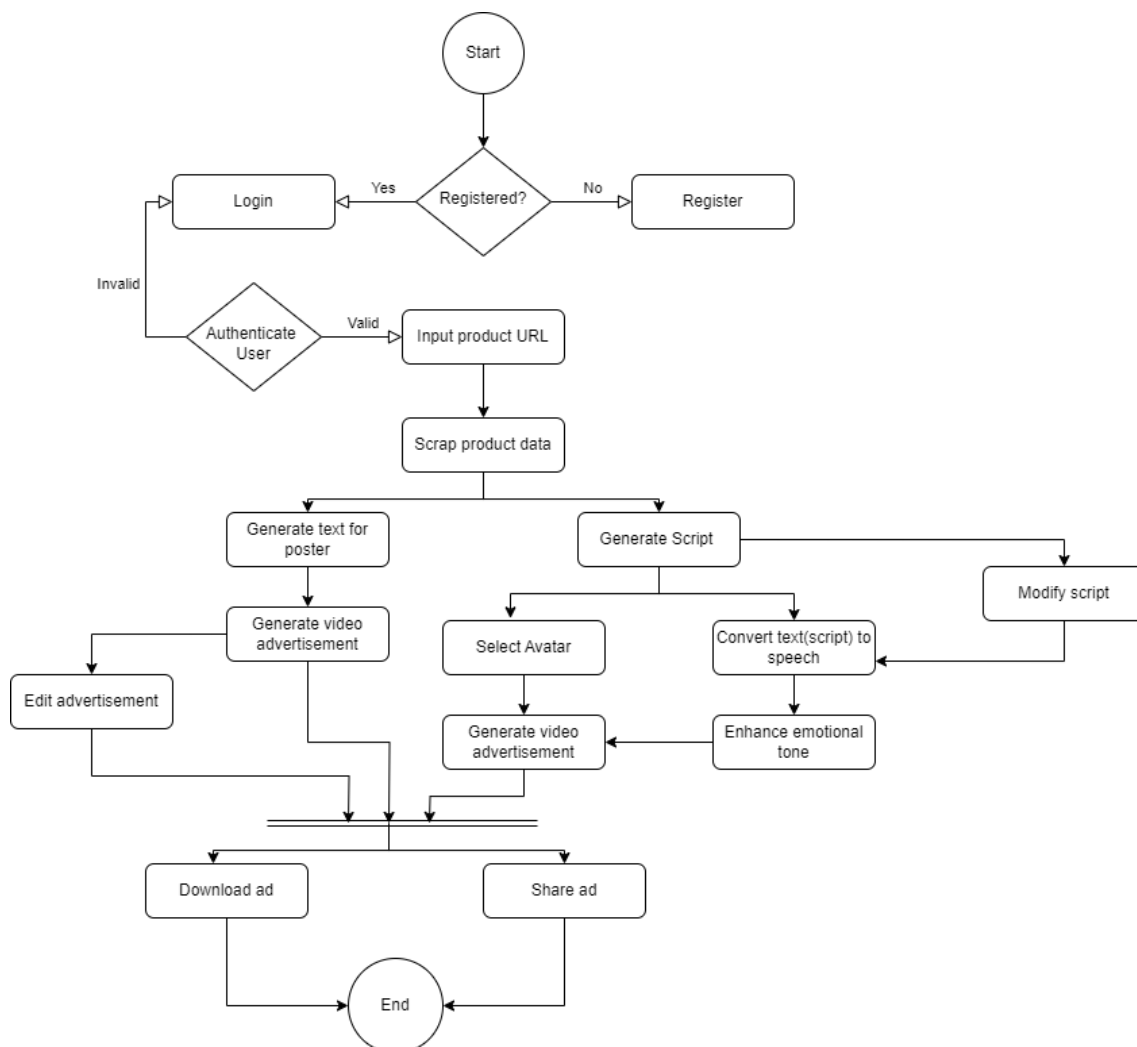


Figure 3.3: Activity Diagram

3.2.2 Data Flow Diagram

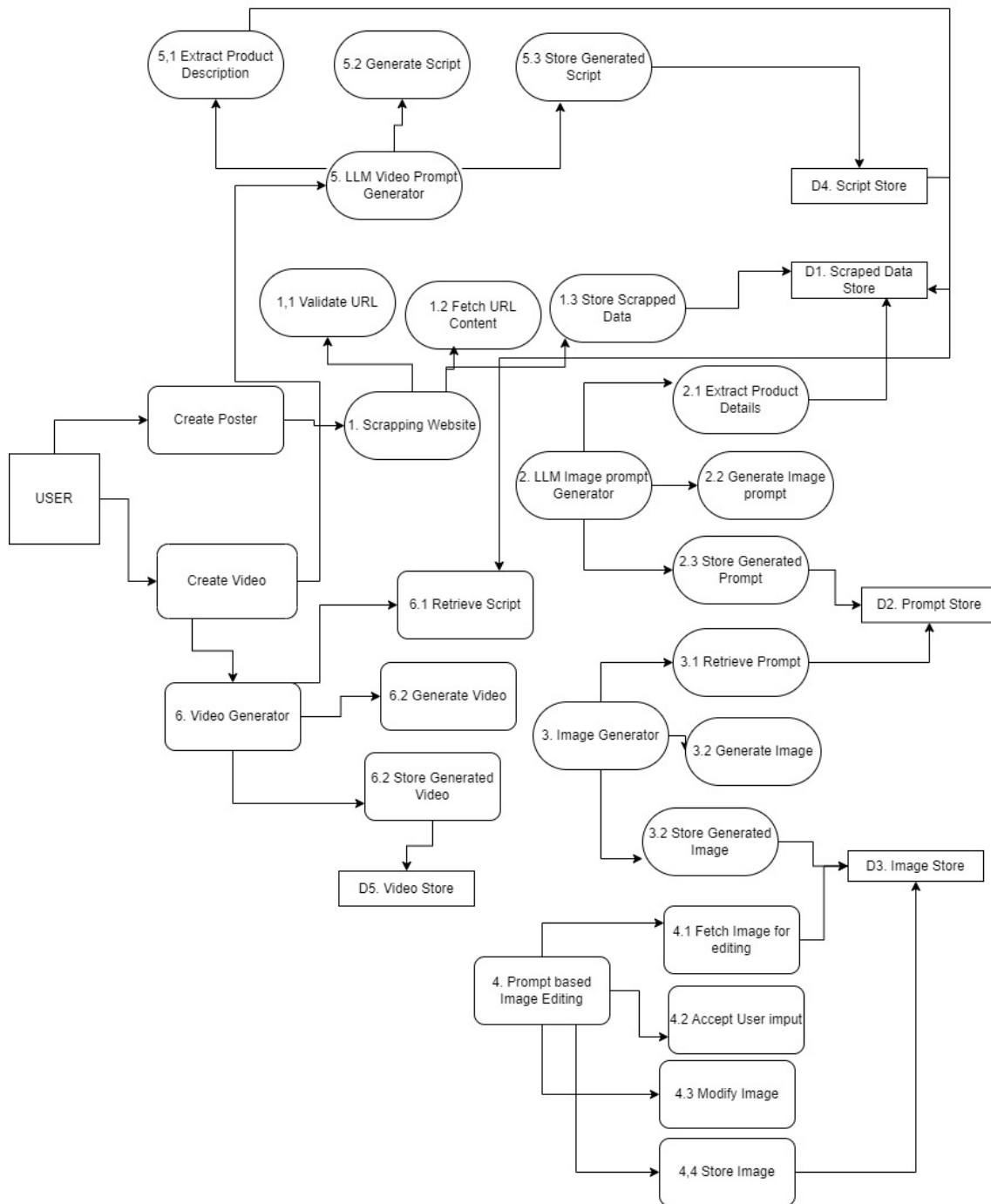


Figure 3.4: Data Flow Diagram

3.2.3 System-level Sequence Diagram

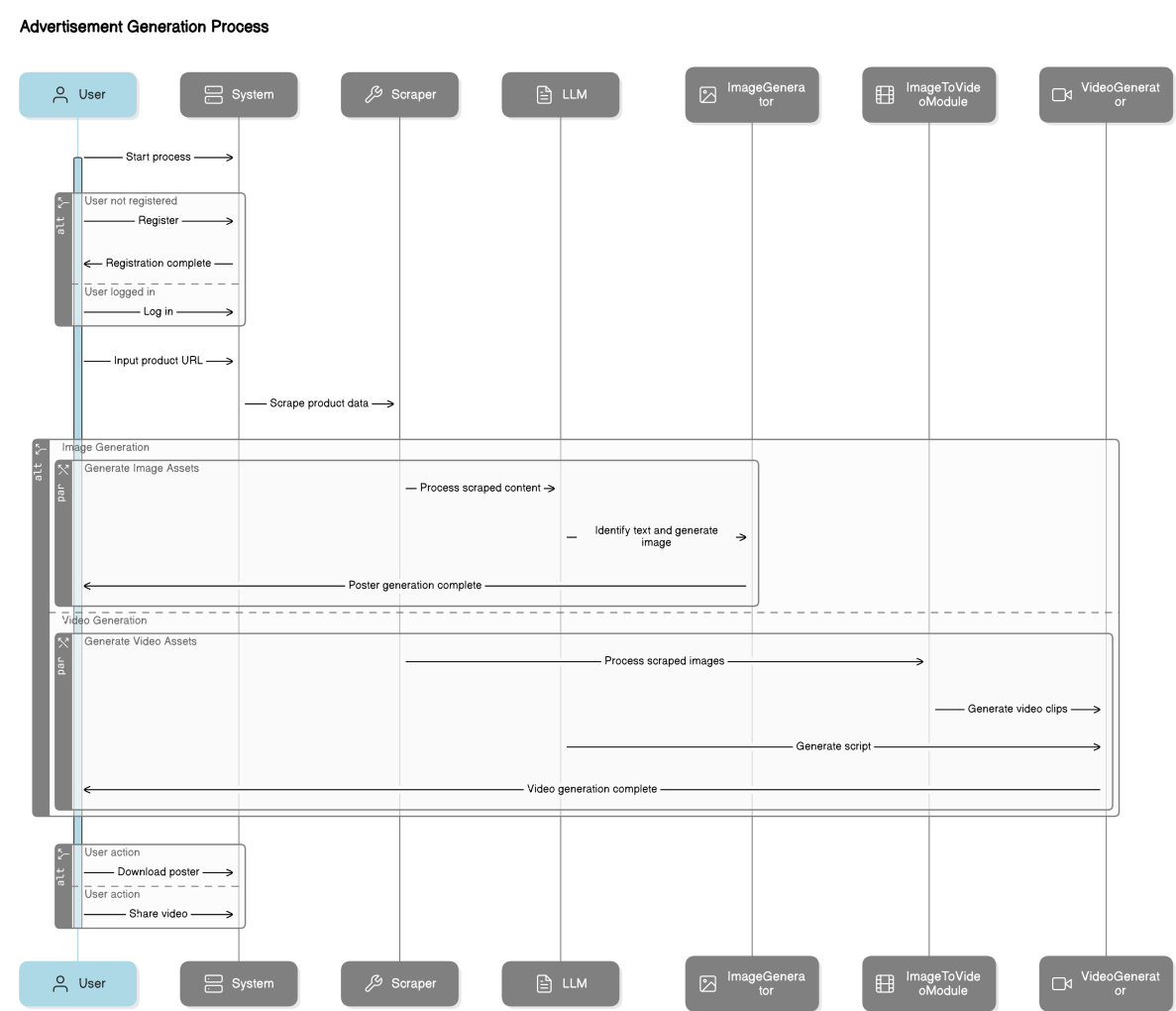


Figure 3.5: Activity Diagram

3.2.4 State Transition Diagram

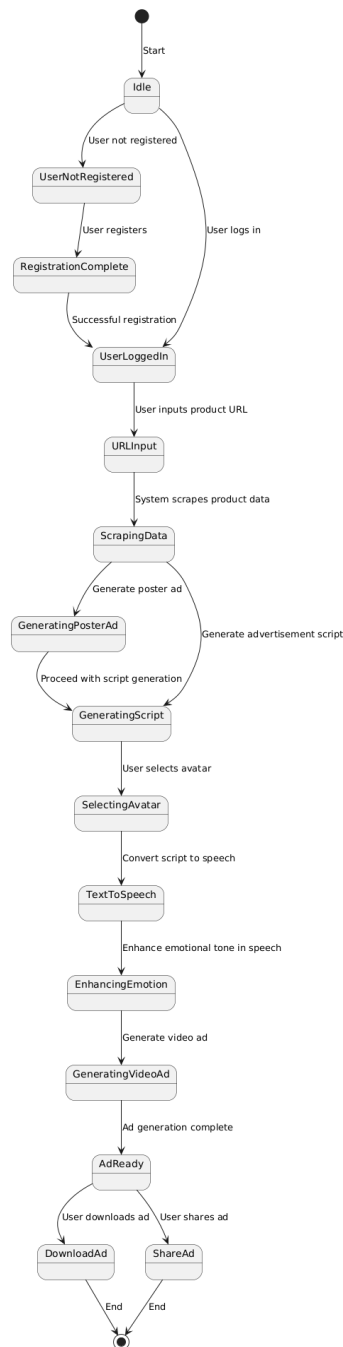


Figure 3.6: Activity Diagram

3.3 Data Design

This section describes the application's data storage and updating procedures. I'll walk through key entities and their operations.

3.3.1 Roles and Users

3.3.1.1 Roles Table

Contains only two records: one for regular users and one for administrators. It contains role information like "admin" or "user."

3.3.1.2 Users Table

Holds user information, including `user_id`, `name`, `email`, `password`, and `role_id`, which identifies each user's unique role.

Storage Example

```
{
  "user_id": 123,
  "name": "Ibrahim",
  "email": "ibrahim@example.com",
  "password": "hashed_password",
  "role_id": "admin"
}
```

Updates When a user registers, their data is inserted into the users table. Admins can update user roles by changing the `role_id` in the users table.

3.3.2 Projects

3.3.2.1 Projects Table

Stores project information for both image and video projects. Important fields include `project_name`, `project_type` (image or video), `status`, and `timestamps`.

Storage Example

```
{
  "project_id": 12,
  "project_name": "Chair Ad for interwood",
  "user_id": 123,
  "created_at": "2024-10-06T12:34:56",
}
```

```

    "updated_at": "2024-10-06T13:45:00",
    "project_type": "video",
    "status": "completed"
  }

```

Updates When a project is created, the data is inserted into the projects table. Project status is updated as the workflow progresses (e.g., from 'in progress' to 'completed'). Whenever a user works on the project, the last update time is also updated.

3.3.3 Image Projects

3.3.3.1 Image Projects Table

It holds information unique to projects involving images. Fields include text (the one that is to be displayed on the poster), base_image_path (the path where the image is stored in the cloud), and generated_image_path (the path where the generated image is stored). Keeping the original information is mandatory, as this allows the user to work on a previous project and update it.

Storage Example

```

{
  "image_project_id": 122,
  "project_id": 11,
  "text": "Buy 1 get 1 Free",
  "base_image_path": "/images/base_image1.png",
  "generated_image_path": "/images/generated_image1.png"
}

```

Updates When an image project is created, details like the base image and text are saved. The generated image is saved after processing, and the generated image path is updated. This allows the user to access previous projects and even make changes.

3.3.4 Video Projects

3.3.4.1 Video Projects Table

Stores details specific to video-based projects. Fields include script_text, voice_path, final_video_path, and the selected avatar_id.

Storage Example

```
{
  "video_project_id": 12,
  "project_id": 11,
  "avatar_id": 0,
  "script_text": "Welcome back to the new video yxz",
  "voice_path": "/voices/voice1.mp3",
  "final_video_path": "/videos/final_video1.mp4"
}
```

Updates The user chooses an avatar, and the avatar id is chosen. script text and voice path are updated once the script is generated and converted to speech. The final video path is updated once the final video is created.

3.3.5 Avatars

3.3.5.1 Avatars Table

Stores the available avatars that users can select for video projects.

Storage Example

```
{
  "avatar_id": "a1",
  "avatar_name": "John",
  "avatar_path": "/avatars/avatar1.svg"
}
```

Updates There will be just two avatars in our project.

3.3.6 Scraped Images and Generated Videos

3.3.6.1 Scraped Images Table

Saves images that have been scraped from the input URL for use in video and image projects.

Storage Example

```
{
  "image_id": 2,
  "project_id": 1,
  "image_path": "/scraped_images/image1.png"
}
```

3.3.6.2 Generated Videos Table

Stores individual video clips generated from scraped images, later joined into a final video.

Storage Example

```
{
  "video_id": 1,
  "project_id": 1,
  "video_path": "/generated_videos/video1.mp4"
}
```

Updates Scraped images are stored immediately after scraping. Generated videos are stored as parts of the video project are completed and then joined into a final video.

3.4 Summary of Data Flows

When the user Creates an account The data is stored in the users tables, when he/she enters a URL into the field, both the text and images are scrapped from the URL and saved, the text is saved in database, while the images are saved in cloud storage and then their paths are saved in the database. The next step is to create either a video or an image; once that is created, we save the generated image or videos depending on the the type and they are stored in their respective tables. this is how the data is majorly stored

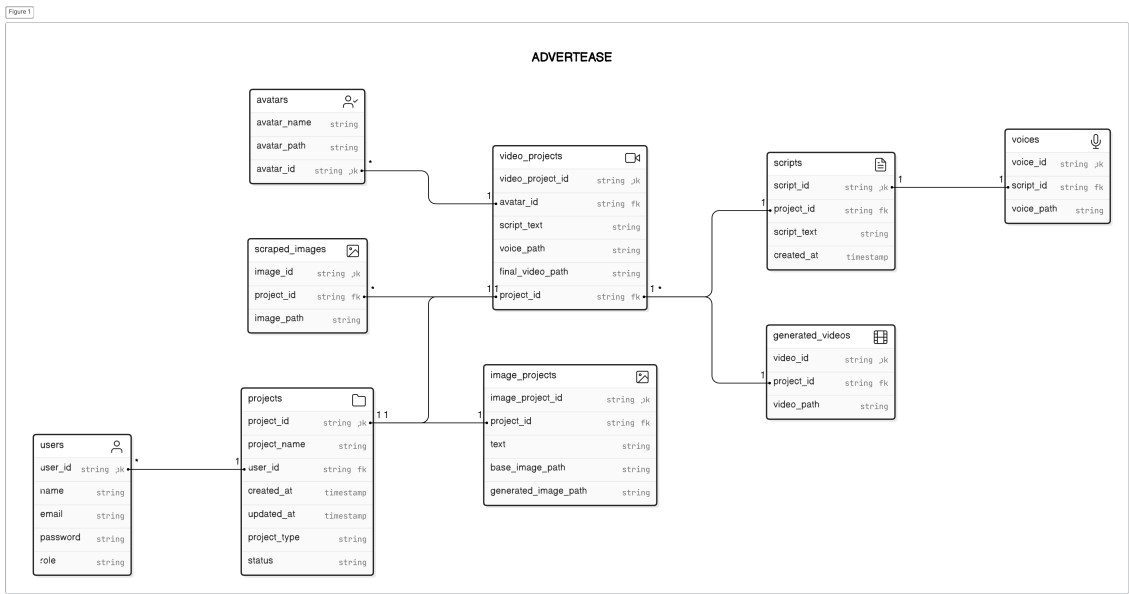


Figure 3.7: Activity Diagram

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